

Data Science

Lecture 09: Association Rule Mining

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Module 1

Itemsets and Their Search Space

***Guiding Question:** Ten thousand products give 2^{10000} candidate itemsets. How can this problem possibly be solvable?*

The Question

Given a set of transactions — shopping baskets, clicked pages, co-occurring symptoms — which **items appear together** more often than chance, and which **imply** which?

Support

$$P(A \cap B) > \sigma$$

How often do these items occur in the same transaction?

Confidence

$$P(A \mid B) > \gamma$$

Given B is in the transaction, how often is A too?

Both are quantities the students already know: support is a **joint** probability, confidence a **conditional** one, both estimated by counting transactions.

Formalisation

Let I be a set of **items** and X a set of **transactions**, each $x \subseteq I$.

- An **itemset** is any $A \subseteq I$ a **k -itemset** has $|A| = k$.
- Its **support** is $\sigma(A) = \frac{|\{x \in X : A \subseteq x\}|}{|X|}$.
- A is **frequent** if $\sigma(A) \geq \sigma_{\min}$.

The Search Space

All itemsets over I form a **lattice** ordered by inclusion — with $2^{|I|}$ elements.

A supermarket with 10,000 products has 2^{10000} candidate itemsets. Enumeration is impossible; the entire chapter is about avoiding it.

Support and confidence are joint and conditional probabilities estimated by counting. The difficulty is not the measure but the **size of the search space**.

Module 2

The A-Priori Algorithm

Guiding Question: *What single structural fact about support lets us discard most of an exponential space without looking at the data?*

The Property That Makes It Possible

A-priori property (anti-monotonicity). Every subset of a frequent itemset is frequent — equivalently: if an itemset is **infrequent**, **every superset of it is infrequent too**.

Adding an item can only ever *reduce* the number of transactions containing the set. So an infrequent itemset prunes its entire superset subtree from the search.

The Algorithm

A-Priori

L^1 = all frequent 1-itemsets.

for $k = 2$ $L^{k-1} \neq \emptyset$ $k++$: **generate** candidates $C_k = \{a \cup b \mid a \in L^{k-1}, b \in L^1, b \not\subseteq a\}$
prune any candidate having an infrequent subset **count** the support of the survivors in one pass over the transactions $L^k = \{c \in C_k \mid \sigma(c) \geq \sigma_{\min}\}$

Generate

build k -candidates from frequent $(k-1)$ -itemsets

Prune

discard candidates with any infrequent subset — no data access needed

Count

one database pass per level

Cost and Improvements

What it costs

One pass over the transactions **per level k** , plus candidate generation. The bottleneck is I/O and the number of candidates.

How it is improved

Hashing candidates into buckets, transaction reduction (dropping transactions containing no frequent itemset), partitioning and sampling.

Mini-Exercise

$\{\text{bread, milk}\}$ has support 0.18, and $\sigma_{\min} = 0.22$.

1. Is $\{\text{bread, milk}\}$ frequent?
2. What can you say about $\{\text{bread, milk, eggs}\}$?
3. How many database passes did answering 2 require?

Mini-Exercise — Solution

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

Mini-Exercise — Solution

1. No — $0.18 < 0.22$.

2. It is **infrequent**, necessarily: its support cannot exceed that of any subset.

3. **None.** The conclusion follows from the property alone. That is exactly where the savings come from.

A-Priori turns one structural fact — **infrequent sets have no frequent supersets** — into a level-wise search that prunes most of an exponential space without touching the data.

Module 3

From Itemsets to Rules

Guiding Question: *A rule holds 80% of the time — and tells you nothing. How is that possible, and what should you measure instead?*

Generating Rules

An **association rule** $A \Rightarrow B$ (with $A \cap B = \emptyset$) has

$$\text{support} = \sigma(A \cup B), \quad \text{confidence} = \frac{\sigma(A \cup B)}{\sigma(A)}$$

Every frequent itemset F is split in all possible ways into A and $B = F \setminus A$ rules meeting the confidence threshold are kept.

Support comes from the itemset and is already known. Only confidence must be computed — so rule generation needs **no further database passes**.

Confidence Is Not Enough

Suppose 80% of all transactions contain milk. A rule $\text{bread} \Rightarrow \text{milk}$ with **confidence 0.80** has learned nothing: milk appears in 80% of baskets regardless of bread.

$$\text{lift}(A \Rightarrow B) = \frac{P(B \mid A)}{P(B)} = \frac{\sigma(A \cup B)}{\sigma(A)\sigma(B)}$$

lift > 1

A makes B **more** likely — a positive association.

lift = 1

A and B are **independent** — the rule carries no information.

lift < 1

A makes B **less** likely — the rule holds in reverse.

Lift is the independence test from Lecture 02 written as a ratio: it compares the joint probability against the product of the marginals.

Mini-Exercise

A retailer reports: nappies $\Rightarrow$ beer, support 0.02, confidence 0.60. Beer appears in 55% of all transactions.

1. Compute the lift.
2. Is the rule actionable?
3. What would make it more convincing?

Mini-Exercise — Solution

Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors.

Mini-Exercise — Solution

1. $\text{lift} = 0.60/0.55 \approx 1.09$ — barely above 1.

2. Hardly. A 9% relative increase over the base rate, on 2% of transactions, is a weak effect on a small slice.

3. A higher lift, a larger support, and evidence that it **persists** — across stores, seasons and time. And a mechanism: a rule with no explanation is a correlation.

Confidence alone rewards **popular** consequents. **Lift** corrects for the base rate by comparing against independence — the same correction Bayes' theorem makes for priors.